

Project data dictionary

Variables, units, file sizes, and controlled data-quality features for the MATH 346 synthetic building-energy projects.

Scenario and unit of analysis

Each file describes a different **synthetic university teaching building**. One row represents one calendar day. The values are simulated for this course; they are not observations from an actual campus building or actual people.

All teams receive the same variables and a comparable amount of missingness, noise, and unusual behavior. Building characteristics, weather realizations, model coefficients, and numerical answers differ by group.

Project 1: MATLAB file

Each file contains 121 daily records from 1 January through 30 April 2024. One heating-energy value is missing, and a small number of observations are deliberately unusual. Handle these features through code rather than editing the CSV manually.

Variable	Type	Unit or values	Meaning
observation_id	text	none	Unique record identifier, such as P1-001
date	date	YYYY-MM-DD	Calendar date of the daily observation
outdoor_temp_c	numeric	degrees C	Simulated mean outdoor temperature
heating_energy_kwh	numeric	kilowatt-hours	Simulated daily heating-energy consumption; one value is missing

The accompanying assignment file contains the group identifier and the temperature at which the fitted model must make a prediction. The assigned temperature lies inside the observed temperature range.

Project 2: R training file

Each training file contains 301 rows representing 300 distinct days from 1 January through 27 October 2025. It contains one repeated record and a small, controlled number of missing values. Identifying and handling these records is part of the required R workflow.

Variable	Type	Unit or values	Meaning
observation_id	text	none	Daily record identifier; unique after cleaning
date	date	YYYY-MM-DD	Calendar date of the daily observation
outdoor_temp_c	numeric	degrees C	Simulated mean outdoor temperature
occupancy_count	numeric	people	Simulated daily building occupancy
wind_speed_m_s	numeric	metres/second	Simulated mean wind speed
day_type	category	weekday, weekend	Building operating-day category
heating_energy_kwh	numeric	kilowatt-hours	Simulated daily heating-energy consumption

Project 2: R validation file

Each validation file contains 65 complete daily records from 28 October through 31 December 2025. It uses the same variables and units as the training file and contains one row for every required prediction.

The validation file is released only after the group has submitted its frozen model plan. Apply the recorded model to these observations, calculate the required metrics, and save one prediction per observation_id.

Interpretation boundaries

- The data support association and prediction, not causal claims.
- The response is heating energy, not total building electricity use.
- Occupancy values are simulated counts and contain no personal information.
- A model that fits one synthetic building need not transfer to another.
- Unusual observations should be investigated and included in a documented sensitivity analysis; do not delete them merely because they are inconvenient.